EL MOUATAMID Abdellatif
Junior Professor Chair, Espace-Dev Research Unit, University of French-Guiana
Ph.D. in Energy Engineering and Automation

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Biography

Elmouatamid Abdellatif is a Junior Professor Chair at the University of French Guiana, France. He was a Postdoctoral research associate at the Department of Electrical and Computer Engineering, New Jersey Institute of Technology, USA (2022-2024). He was in a postdoctoral position at IRESEN (Institute of Research in Solar Energy and New Energies), Morocco (2021-2022). He received a Ph.D. in Energy Engineering and Automation from Chouaib Doukkali University of El-Jadida, Morocco (Dec-2020). He received a Master from Chouaib Doukkali University with 1st class honors (2016). He has research experience in transmission and distribution lines, power systems, energy efficiency in buildings, smart microgrids, energy management, anomaly detection, and predictive maintenance, and engaged in numerous collaborative research projects.

Professional Experience

2024-Present	Junior Professor Chair at the Department of Sciences and Technology, University of French-Guiana, Espace-Dev Research Unit, France.
2022-2024	Post-Doctoral Research Associate at the Department of Electrical and Computer Engineering, New Jersey Institute of Technology, United States.
2021-2022	Post-Doctoral Researcher, IRESEN (IRESEN-Solar Energy and New Energies Research Institute), Morocco.

Education

2017-2020	Ph.D. in Energy Engineering and Automation, Chouaib Doukkali University, Faculty of Sciences, Morocco (December 2020).
2014-2016	Master's degree in industrial Electronics & Telecommunications, Chouaib Doukkali University, Faculty of Sciences, Morocco.
2010-2014	Bachelor's degree in information and communication technologies, Chouaib Doukkali University, Faculty of Sciences.
2010	Baccalaureate diploma in sciences.

Editorial Activities and Committees

2023-Present	Guest editor (Special Collection): Fault Detection, Classification, and Diagnostic in Microgrid Systems Infrastructure Using Data Mining Approaches, Journal of Measurement and Control.
2022-Present	Editorial Board Membership of International Journal of Sensors and Sensor Networks (IJSSN) ISSN Print: 2329-1796; ISSN Online: 2329-1788.
2022-Present	Associate member in STIC-Lab (Technological Systems of Information and Communication Laboratory), Chouaib Doukkali University, Faculty of Sciences, Morocco.
2021-Present	Volunteer Reviewer Board of Multidisciplinary Digital Publishing Institute (MDPI) journals: Energies, Electronics, Sensors, Smart Cities.
2020-Present	Volunteer Reviewer Board of IEEE Access Journal.

2023	Guest editor (Special Issue): Latest Developments in Offshore Wind Technologies, Energies Journal.
2023	Bootcamp Organizer: New Jersey 2-Day Offshore Wind Bootcamp Training, New Jersey Institute of Technology, USA.
2023	Community member: GREEN-2022 and GREEN-2023, The Eighth International Conference on Green Communications, Computing and Technologies, Portugal.
2022	Speaker/Organizer: North Jersey Research Student Conference 2022, New Jersey Institute of Technology, USA.
2020	Technical Program Committee Member, The International Workshop on Emerging Networks and Communications (IWENC 2019), Nov 2020, Coimbra, Portugal.
2019	Community member of the 7th Job-Fair event at the International University of Rabat hosted many stands for more than 35 companies from different fields.

Relevant Training & Experiences

Mar 2023	OPAL-RT: Hardware-in-the-Loop and Real-Time Simulation Training.
Dec 2019	Internship: UMI-GT-CNRS (*Unité Mixte Internationale-Georgia Technologie-Centre National de la Recherche Scientifique*) in Georgia Tech Lorraine, France. Topic: The development of photovoltaic energy research to establish protocols, and tests in common between GT-CNRS and the UIR (International University of Rabat).
Nov 2019	Workshop trainer: IoT/Big-Data technologies for embedded systems, ENSA (National School of Applied Sciences) Marrakech.
Aug-Sept 2019	Participant in the international competition Solar Decathlon AFRICA: Renewable energy sources and data monitoring platforms installation, held in Green Energy Park, Benguerir, Morocco.
Dec 2018	TRNSYS Certificate.
Oct 2018	Raspberry-Pi, Arduino Developer Certificate.
Oct 2018	Patent writing Certificate, OMPIC (Office Marocain de la Propriété Industrielle et Commerciale).
Mar 2018	Training workshop: Patent conscripting and drafting, OMPIC.
Jul 2017	Training: Course LabVIEW, Core 1 and Core 2.
Apr-Jun 2016	Internship: Study of Industrial Sensors for surveillance, within the College of Sciences in El-Jadida, STIC laboratory (*Systèmes Technologiques de l'information et de communication*).
May 2015	Microsoft Certified Solutions Associate: Windows Server 2012, Server Infrastructure.
Mar 2013	Certificate in Cisco Certified Network Associate (CCNA).
Aug 2013	Initiation Internship within OCP (Office Chérifien des Phosphates) Group in Jorf-Lasfar: Study of the existing Network in the region of Jorf-Lasfar and Migration to IPV6 using GNS3, VMWARE WORKSTATION, PAQUET TRACER, and Wireshark.

Synergistic Activities

THyGRE
2024-2029
ANR-UG

- Project Manager: Hydrogen Transition Applied to Electrical Grid Management in Non-Interconnected Areas.

The research project aims to develop expertise on the site based on two objectives: (i) to develop innovative methods for managing multi-source systems; (ii) to characterize the services that hydrogen storage can provide to off-grid networks.

HOLSYS
2019-2021
UIR-IRESEN

- Postdoc member of HOLSYS Project: UIR-IRESEN-ENSAK.

The project developed a Software/Hardware platform for real-time data collection and communication interfaces in Microgrid systems for both energy and comfort management.

MIGRID
2017-2020
UIR- UCD

- Ph-D Student member of MIGRID Project: USAID-PEER Project.

The project developed new approaches for Renewable Energy Sources integration in the buildings with fuel cells and battery storage system Energy Efficiency.

Solar Decathlon AFRICA
Aug-Sep 2019
UIR-IRESEN

- Ph-D Student member of Solar Decathlon AFRICA Project, IRESEN, Morocco.

The project challenges collegiate teams to design and build houses powered exclusively by the sun, to increase energy efficiency, and to construct buildings with local materials.

Publications

Journals Papers:

[1] **Elmouatamid**, A., Chowdhury, M. A., Netto, M., & Pong, P. W. (2024). ''Economic feasibility study of standalone community microgrid in 37 cities of the USA'', HKIE Transactions, Volume 31 - Issue 1, p. 1-17.

[2] **Elmouatamid**, A., Brian F., Jian S., and Philip W. T. P. (2023). "Air Conditioning Systems Fault Detection and Diagnosis-Based Sensing and Data-Driven Approaches" Energies 16, no. 12: 4721.

[3] **Elmouatamid**, A., Ouladsine, R., Bakhouya, M., El Kamoun, N., & Zine-Dine, K. (2021). "*A Predictive Control Strategy for Energy Management in Micro-Grid Systems"*. Electronics.

[4] **Elmouatamid**, A., Ouladsine, R., Bakhouya, M., El Kamoun, N., Zine-Dine, K., & Khaidar, M. (2020). "*Review of Control and Energy Management Approaches in Micro-Grid Systems"*. In Smart Grids and Microgrids. Energies.

[5] **Elmouatamid**, A. (2020). MAPCAST: "A*n Adaptive Control Approach using Predictive Analytics for Energy Balance in Micro-Grid Systems"*. International Journal of Renewable Energy Research (IJRER), 10(2), 945-954.

[6] **Elmouatamid, A**., NaitMalek, Y., Bakhouya, M., Ouladsine, R., Elkamoun, N., Zine-Dine, K., & Khaidar, M. (2019). "*An energy management platform for micro-grid systems using Internet of Things and Big-data technologies"*. Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering, 0959651819856251.

[7] Raj, R., **Elmouatamid**, A., Swain, A., & Pong, P. W. (2025). Critical review of hybrid modular multilevel converters for HVDC transmission in offshore wind energy. Energy Reports, 14, 1761-1778.

[8] Elkhoukhi, H., **Elmouatamid**, A., Haibi, A., Bakhouya, M., & El Ouadghiri, D. (2025). Overview of Sensing and Data Processing Technologies for Smart Building Services and Applications. Sustainability, 17(9), 4029.

[9] Elkhoukhi, H., Bakhouya, M., **Elmouatamid**, A., El Ouadghiri, D. and Gaber, J., 2025. A review of occupancy sensing technologies and approaches in smart buildings. *International Journal of RF Technologies*, *15*(1), pp.19-48.

[10] Akhyurna, S., **Elmouatamid**, A., Ahmed, M., Philip W.T. (2022). "*Sensor Technologies for Transmission and Distribution Systems: A Review of the Latest Developments. In Condition Monitoring of Critical Infrastructure for Energy Systems"*, Energies-1917449.

[11] Elouadoud, H., Ouladsine, R., **Elmouatamid**, A., Bakhouya, M., & Pierfederici, S. (2022). "*Towards an Advanced Control Approach for Energy Management in Distributed Micro-Grid Systems"*. Available at SSRN 4191086.

[12] Alidrissi, Y., Ouladsine, R., **Elmouatamid**, A., Errouissi, R., & Bakhouya, M. (2022). "*Constant Power Load Stabilization in DC Microgrids Using Continuous-Time Model Predictive Control".* Electronics, 11(9), 1481.

[13] Alidrissi, Y., Ouladsine, R., **Elmouatamid**, A., & Bakhouya, M. (2021). "*An Energy Management Strategy for DC Microgrids with PV/Battery Systems"*. Journal of Electrical Engineering & Technology, 1-12.

[14] S. Boulmrharj, Y. NaitMalek, **A. Elmouatamid**, M. Bakhouta, R. Ouladsine, K. Zine-dine, M. Khaidar, M. Siniti. "*A Dimensioning Approach for Micro-Grid Systems"*. Energies Journal 2018.

Chapter Book:

[15] **Elmouatamid A.** et al. (2021) "*Towards an IoT/Big-Data Platform for Data Measurements, Collection and Processing in Micro-grid Systems"*. In: Motahhir S., Bossoufi B. (eds) Digital Technologies and Applications. ICDTA 2021. Lecture Notes in Networks and Systems, vol 211. Springer, Cham. https://doi.org/10.1007/978-3-030-73882-2_26

[16] **Elmouatamid**, Abdellatif, et al. "*A microgrid system infrastructure implementing IoT/Big-Data technologies for efficient energy management in buildings*." Advanced Technologies for Solar Photovoltaics Energy Systems. Cham: Springer International Publishing, 2021. 571-600.

Patents:

[17] **A. Elmouatamid,** Y. NaitMalek, S. Boulmrharj, M. Bakhouya, R. Ouladsine, M. El kamoun, K. Zine-dine, M. Khaidar "*Procédé de monitoring, estimation, prédiction et contrôle en temps réel des systèmes multi-sources*", National publication No: MA 43611 A1, International CI: G01R 21/00; G01R 21/00

Conferences:

[18] Kabir, R., **Elmouatamid**, A., Elkhoukhi, H., & Pong, P. W. (2024, February). Photovoltaic Power Forecasting Using Neural Networks for Short and Medium-Term Dependencies. In 2024 IEEE Texas Power and Energy Conference (TPEC) (pp. 1-6). IEEE.

[19] Swain, Akhyurna, **Elmouatamid** Abdellatif, and Philip WT Pong. "Magnetic Signature-Based Model Using Machine Learning for Electrical and Mechanical Faults Classification of Wind Turbine Drive Trains." 2024 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT). IEEE, 2024.

[20] **A. Elmouatamid,** R. Ouladsine, M. Bakhouya, N. Elkamoun, K. Zine-Dine, M. Khaidar, "*A Control Strategy Based on Power Forecasting for Micro-Grid Systems*", 5th IEEE International Smart Cities Conference'2019, Casablanca, Morocco.

[21] **A. Elmouatamid**, R. Ouladsine, M. Bakhouya, N. El Kamoun, K. Zine-Dine and M. Khaidar, "*A Model Predictive Control Approach for Energy Management in Micro-Grid Systems*" *2019 International Conference on Smart Energy Systems and Technologies (SEST)*, Porto, Portugal, 2019, pp. 1-6.

[22] **A. Elmouatamid,** R. Ouladsine, M. Bakhouya, M. El kamoun, K. Zine-dine, M. Khaidar, R. Abid, "*Deployment and Experimental Evaluation of Micro-Grid Systems*", the 6th International Renewable and Sustainable Energy Conference (IRSEC'2018), Rabat, Morocco.

[23] **A. Elmouatamid,** Y. NaitMalek, R. Ouladsine, M. Bakhouya, K. Zine-dine, M. Khaidar, M. Elkamoun, R. Abid "*Towards a Demand/Response Control Approach for Micro-grid Systems*" 2018 5th International Conference on Control, Decision and Information Technologies (CoDIT). IEEE, 2018.

[24] **A. Elmouatamid,** R. Ouladsine, V. Felix, M. Bakhouya, K. Zine-dine, M. Khaidar, M. Elkamoun, R. Abid, "*Modeling and Performance Evaluation of Photovoltaic Systems*", IRSEC'2017, IEEE, Tanger, Morocco.

[25] **A. Elmouatamid,** N. Elkamoun, F. Lakrami, M. Bakhouya, "*Wireless Sensor Networks for Smart Micro-Grids*", CMT'2018, El jadida, Morocco.

[26] M. ElIbrahimi**, A. Elmouatamid**, M. Bakhouya, K. Feddi, R. Ouladsine "*Performance Evaluation of Fixed and Sun Tracking Photovoltaic Systems*", in the 6th International Renewable and Sustainable Energy Conference (IRSEC'2018), Rabat, Morocco.

[27] M. Bakhouya, Y. NaitMalek, **A. Elmouatamid**, F. Lachhab, A. Berouine, S. Boulmrharj, R. Ouladsine, V. Felix, K. Zine-dine, M. Khaidar, N. Elkamoune, "*Towards a Data-Driven Platform using IoT and Big Data Technologies for Energy Efficient Buildings*", in CloudTech'2017, pp. 1-5, Rabat, Morocco

[28] Boulmrharj, S., NaitMalek, Y., **Elmouatamid, A**., Bakhouya, M., Ouladsine, R., Zine-Dine, K., ... & Abid, R. (2018). "*Approach for dimensioning stand-alone photovoltaic systems"*. Energy Procedia, 153, 56-61.

[29] S. Boulmrharj, Y. NaitMalek**, A. ElMouatamid,** , R. Ouladsine, M. Bakhouya, M. Ouldmoussa, K. Zine-dine, M. Khaidar, R. Abid, "*Towards a Battery Characterization Methodology for Performance Evaluation of Micro-Grid Systems"*, International Conference on Smart Energy Systems and Technologies, Sevilla, Spain, 2018.

Teachings

Years - Level		Institution	Course	Hours
2024-2025	Master	University of French Guiana Master: Energy	Energy sources and storage systems	42
2024-2025	Master	University of French Guiana Master: Energy	Artificial intelligence for power systems	42
2024-2025	Master	University of French Guiana Master: Energy	Energy conversion	42
2024-2025	Master	University of French Guiana Master: Energy	Electrical Machines	42
2024-2025	Master	University of French Guiana Master: Energy	Research/innovation projects supervising	42
2023-2024	Master	ECE 449 - NJIT Master: Electrical system and Energy production	Energy management	24
			Electricity transport and distribution systems	24
2021-2022	Engineering school	ECINE 3-S5 International University of Rabat	Control Signals	20
			Electronic Signals and Systems: Signal Processing and Conception	20

2020-2021	Engineering school	ECINE 3-S4 International University of Rabat	Electricity	32
2020-2021	Engineering school	ECINE 4-S8 International University of Rabat	Renewable Energy Sources	36
2020-2021	Engineering school	ECINE 3-S5 International University of Rabat	Automatic: Control & Command	16
			Electronic Signals and Systems: Signal Processing and Conception	32
2020-2021	Engineering school	CPI 2-S4 International University of Rabat	Signals and Systems	40
2019-2020	Engineering school	TICLab 2-S3 International University of Rabat	Computer Networks	16
2019-2020	Engineering school	ECINE 3-S5 International University of Rabat	Automatic: Control & Command	16
			Electronic Signals and Systems: Signal Processing and Conception	32
2018-2019	Engineering school	ECINE 2-S4 International University of Rabat	Automatic: Control & Command	32
2018-2019	Bachelor's degree	Master Chouaib Doukkaly University	Computer Networks	40
2017-2018	Bachelor's degree	Bachelor's degree Chouaib Doukkaly University	Telecommunications, Computer Networks	56
2016-2017	Bachelor's degree	Bachelor's degree Chouaib Doukkaly University	Office automation software Literature research Graduation project redaction	60
Total				**701**

Co-supervision

- **PhD student:**

  Swain, Akhyurna, "*Condition Monitoring and Fault Detection of Offshore Wind Turbine Drive Train*", Department of Electrical and Computer Engineering, New Jersey Institute of Technology, USA (2022-2025).

  Alidrissi Youssef, "*A Model Predictive Control Strategy for Energy Management in Microgrid System Using Machine Learning Algorithms*". LERMALab, Rabat, International University of Rabat (2020-2024).

- **Undergraduate and graduate students:**
  - ✓ Supervision of master's students in work-study projects in collaboration with companies (France),
  - ✓ Semestrial supervision of undergraduate and graduate students' projects (from 2022 to the present day),
  - ✓ Supervising more than 20 students for graduation projects and mini projects (2017-2022) at the International University of Rabat,
  - ✓ Participating with students in national and international computations about innovation and entrepreneurship.

Awards & Honors

Oct-2022	Postdoc of the Month: Department of Electrical and Computer Engineering, New Jersey Institute of Technology, United States.
2019	Best Poster Award: LERMA (Renewable Energies and Materials) Laboratory Seminar held at the International University of Rabat.
2016	First Class Honors: Master: Industrial Electronics and Telecommunication, Chouaib Doukkali University.

Skills

Computer	Hardware and Power Hardware in the Loop (PHIL): OPEL-RT, RTDS, DSpace.

& Systems	Mastering Simulation tools: MATLAB, PVsyst, RETScreen, EnergyPlus, COMSOL Multiphysics, JMAG-Designer, ISIS Proteus, GNS3, Packet Tracer, and VMWARE WORKSTATION.
	Networks Configuration, Architecture, and study of Networks & Routers Configuration.
	Programming languages Python, Matlab.
	Windows Operating System; Linux.
Big-Data/IoT Technologies	Hadoop cluster installation, Node-RED, Thingsboard, Thingspeak.
	Data collection and analyses, Real-time data monitoring, and control.
Languages	Arabic English : Native. French

Links

Google Scholar	https://scholar.google.fr/citations?user=vPh13ZwAAAAJ&hl=fr
Hal.science	https://hal.science/search/index?q=abdellatif+elmouatamid
ORCID	https://orcid.org/my-orcid?orcid=0000-0001-5499-5865
Scopus	https://www.scopus.com/authid/detail.uri?authorId=57201975143
AD Scientific	https://www.adscientificindex.com/scientist/abdellatif-elmouatamid/4360552

References

Philip W. T. Pong	**Najib Elkamoun**	**Khalid Zine-Dine**	**Mohamed Bakhouya**
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New Jersey Institute of Technology, United States	Faculty of Sciences, Chouaïb Doukkali University, Morocco	Faculty of Sciences, Mohammed V University, Morocco	International University of Rabat, Morocco
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